LedgerLens - Case Study Summary

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This document summarizes the case study prompt, defines key financial and audit terms, and explains what the LedgerLens application does and how the data flows end-to-end.

All figures in LedgerLens are computed deterministically. AI is used only for language and optional PDF structure extraction.

1) Case Study - What is being asked
- Build an AI-enabled data preparation pipeline for audit and assurance engagements
- Inputs per client: General Ledger (GL), Trial Balance (TB), supporting journal PDFs
- Problems: messy data, schema variability, missing data, duplicates, GL and TB mismatches
- Requirements: reliable, auditable outputs; scale across many engagements (e.g., 1,500 clients)

2) Glossary - financial and audit terms
General Ledger (GL): detailed transaction postings (journal lines).
Trial Balance (TB): closing balances by account at period end.
Journal / transaction: postings sharing one transaction_id that should net to ~0 (double-entry).
Debit (DR) / Credit (CR): posting direction; two sides of double-entry.
Signed amount: normalized value so sums work. Convention: +amount for DR, -amount for CR.
GL-derived balance: sum of signed amounts by account from the GL detail.
Variance: GL-derived balance minus TB balance for an account.
Break: a non-zero variance (a discrepancy to explain).
Materiality: threshold above which a difference is significant enough to prioritize.
Material break vs immaterial break: variance classification relative to materiality.
Completeness: required fields populated.
Validity: values conform to expected formats/domains.
Consistency: journals balance (double-entry holds).
Uniqueness: no duplicates.
Accuracy (here): GL ties to TB (few break accounts).
Timeliness: dates are parseable and within the reporting period.
Audit trail / lineage: append-only log of transformations and decisions with affected rows.
Supporting documentation: vouchers/approvals in PDFs.
Documentation coverage: share of journals evidenced by the provided PDFs.

3) What the application does
LedgerLens ingests GL and TB exports plus optional supporting PDFs, then cleans and standardizes the data, validates integrity, flags duplicates, reconciles GL versus TB, and produces an audit-ready output package (JSON plus workpaper-style PDF) with a full audit trail.

Deterministic core, AI at the edges

- Deterministic: balances, variances, materiality, KPIs, scorecard, duplicate detection, reconciliation status
- AI (Gemini or OpenAI GPT-4o): explanations and narratives over computed results; optional extraction of voucher fields from PDFs
- Fallback: if AI is unavailable, deterministic templates produce narratives so the run never fails

4) How the data flows (end-to-end)
A) Ingest and schema mapping
- Read CSV/XLSX safely (encoding and delimiter handling). Keep raw values as strings initially.
- Map client-specific headers to a canonical schema via synonym dictionary and fuzzy matching.
- Only unresolved headers may be suggested by AI, and those suggestions are flagged for review.
B) Clean and standardize
- Normalize dates and parse amounts across formats.
- Convert DR/CR to signed_amount for reliable aggregation and reconciliation.
- Produce a cleaning report plus audit trail entries listing what changed and which rows were affected.
C) Validate
- Double-entry: within each transaction_id, sum(signed_amount) should be approximately zero.
- Missing required fields: flagged with row numbers.
- Referential integrity: GL accounts should exist in TB and vice versa (gl_only and tb_only).
- Domain checks: valid DR/CR codes, numeric amounts, dates within period.
D) Deduplicate
- Identify exact duplicates and near duplicates.
- Flag and group; do not delete automatically.
E) Reconcile GL versus TB
- Compute GL-derived balances by account, compare to TB balances, compute variance.
- Classify accounts: reconciled, immaterial_break, material_break, gl_only, tb_only.
F) Supporting documents
- Extract PDF text deterministically and compute documentation coverage.
- Optionally extract voucher metadata using AI, with provenance tags.
G) Outputs
- PipelineResult JSON (single source of truth for UI, reports, and optional Neo4j projection).
- Workpaper PDF and machine-readable JSON for audit evidence and traceability.
- Grounded chat over computed outputs (figures are never invented).

5) Likely interview questions and concise answers
Q: How do you handle data variability?
A: Canonical schema plus synonyms plus fuzzy matching, plus a review workflow for unresolved headers; all decisions are logged.
Q: How do you reconcile GL and TB?
A: Convert DR/CR to signed_amount, sum by account to get GL-derived balances, compare to TB, classify variances using materiality.
Q: Where does AI fit?
A: Only language and optional unstructured extraction; never generates numbers.

Q: How do you ensure auditability?

A: Append-only audit trail with stage, action, rows affected, and provenance tags; export JSON and PDF workpapers.

Q: How does it scale to 1,500 clients?

A: Per-client stateless pipeline jobs; run in parallel; store results per engagement; portfolio view via consistent KPIs.